VOIDSCOPE · DESIGN DECISION RECORD

VoidScope TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	VDS-DDR-001	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/decisions/0001-trl2-review-decisions.md @ 463eaf8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review decisions, decided under Amish's 2026-10-03 pre-approval of the batch and its recommendations

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

The TRL 2 populate step (session 2026-10-03, `docs/REVIEW.md`) answered the open questions of VDS-PRB-001 v0.1 and set the key design choices of VDS-PRC-001 v0.2. Under the normal rules each would be recorded as "Proposed, awaiting Amish". Amish pre-approved every recommendation in this batch run, so each item below is recorded as decided, with its options and the reason for the choice. Choices that touch safety take the conservative option and state what evidence would relax them. Partners and regions are recorded as the first candidate to approach, not as agreed. The TRL 2 approval that the TRL 3 step needs is the same instruction.

Options considered

Table 1. Options for each item.

#	Item	Options	Chosen
D1	Where the water and air lines run	Alongside the rod in clips; inside, each in its own tube; inside, with the rod bore as the air duct and the water in a guide tube	Inside, bore as air duct, water tube in a PTFE guide tube
D2	Rod and joints	Telescopic carbon pole; threaded sections; sections with internal couplers and snap buttons	6061-T6 tube 38.1 x 1.47 mm in four 1,050 mm sections, internal couplers, V-spring snap buttons, lines left threaded through (tent-pole storage)
D3	Head articulation	Articulating head; fixed head aimed by turning the rod	Fixed head, about 90° view, aimed by turning the rod; articulation left for a later version (cost and patent risk)
D4	Camera	Bare board camera in a self-made sealed head; bought sealed pipe-inspection head in a machined nose	Bought IP68 29 mm AHD 1080p head with 12 LEDs, in a turned aluminium nose
D5	Display	Handheld screen on its own cable; monitor in the surface unit lid; phone or tablet over USB	7 in AHD monitor with recorder in the lid of the surface unit (one battery, one cable; USB does not reach 7 m)
D6	Audio	None; separate microphone and speaker; one speaker used as both, with push-to-talk	One 20 mm IP67 speaker at the tip used as speaker and microphone, push-to-talk headset at the surface (fitted, not optional, because talk is how the crew judges consciousness before water)
D7	Water delivery	Open drip tip with a roller clamp; IV-style drip set; gravity drip with a fixed capillary ceiling, roller clamp and a bite valve	Gravity drip: bottle, strainer, 600 mm x 0.5 mm capillary, roller clamp, priming stopcock, single-use tube, bite valve parked in the nose
D8	Air supply	Hand bellows; diaphragm pump with a relief valve; centrifugal blower with shut-off at or below 1 kPa	12 V centrifugal blower chosen by its shut-off pressure, intake filter, flow meter, speed knob
D9	Power	AA cells; power-tool packs; 12.8 V LiFePO4 packs	Two 12.8 V 6 Ah LiFePO4 packs with built-in protection, one fitted and one spare
D10	Tethered crawler block	Share the proposed crawler head interface; stay independent	Independent; no shared head interface, so the RedZone tether odometry question does not arise
D11	Co-design partner and region	Any accredited USAR team; a national fire or civil protection training school in an earthquake-prone country	First candidate to approach: the disaster management training school of the Armed Police Force, Nepal; second: Turkey's AFAD training centres. Not contacted; not partners
D12	Requirements	Keep the TRL 1 set; restate	R2 split into reach (R2) and bends (R11); R5 as a ceiling with a bite valve; R6 adds a blocked-tip limit; R8 as a value-engineering target; R12 two-way voice added
D13	Budget, pitch and problem	Change; keep	Kept: `budget_usd` stays at USD 2,000 as a value-engineering target; pitch and problem unchanged

Decision

All items D1 to D13 are decided as chosen in Table 1, by Amish under his 2026-10-03 pre-approval.

Table 2. Safety choices taken on the conservative side, and what would relax them.

#	Conservative choice	Evidence that would relax it
S1	Drip ceiling sized for a 2 m water height at 40 °C (twice the allowed height), giving 1.5 mL/min at the normal 1 m and 20 °C	Measured flows at the tip with the bought capillary, and the co-design medical lead's view of the rate to offer
S2	Water only through a bite valve, never an open drip tip	A written medical protocol from the co-design partner that allows another mouthpiece
S3	Blower chosen by shut-off pressure (at most 1 kPa), with no relief valve to fail	None needed; a measured blocked-tip pressure at TRL 4 confirms it
S4	Audio fitted as standard, so the crew can judge consciousness before water	The medical lead's view that another check is enough
S5	LiFePO4 packs with built-in protection, charged only away from the kit	None; this is the lowest-risk common lithium chemistry
S6	The probe is never used as a lever or pry bar; the 300 N pull and 200 N push are the design loads	Measured joint and button strength at TRL 4

Consequences

- VDS-PRB-001 v0.2, VDS-PRC-001 v0.2 and VDS-REQ-001 v0.2 carry these decisions; the TRL 3 versions (v0.3) add the constructable design of VDS-DDR-002.
- R11 (bends) is not met by D2 and D3; it is recorded as a requirement not met, not redesigned at TRL 3.
- `project.yaml` : only the TRL fields, `design_state` and the evidence list change. `budget_usd` , the pitch and the problem are unchanged.
- Nothing here authorizes building or testing; TRL 4 is capped.